

Reproducibility Notes

The reproducibility review conducted during this project focused on evaluating whether an independent researcher could reasonably understand and evaluate the dataset using the deposited repository materials and associated publication. Since this project followed a CURATE(D)-based curation approach, the assessment emphasized transparency, documentation quality, and workflow interpretability rather than rerunning statistical analyses.

Reproducibility Strengths

Several aspects of the repository support basic reproducibility and independent access:

- the dataset is publicly available through Dryad,
- the repository provides persistent DOI support,
- the deposited files are accessible in CSV format,
- and the dataset is directly connected to the associated journal article.

These features support discoverability, citation, and access to the shared research materials.

Reproducibility Limitations Identified

During the assessment, several limitations affecting reproducibility were identified.

1. Limited Variable-Level Documentation

Some variables and coded values are not fully explained within the repository documentation, making interpretation difficult without article context.

2. Missing Workflow Transparency

The repository does not clearly document preprocessing activities, scoring calculations, transformation procedures, or intermediate analytical steps.

3. Dependence on the Associated Article

Understanding several behavioral and impulsivity-related measures requires reading the associated publication because methodological explanations are not fully included within the repository itself.

4. Incomplete Transformation Lineage

The dataset appears to include processed or derived analytical variables, but detailed lineage information explaining how these variables were generated is limited.

Impact on Independent Reuse

These limitations do not prevent reuse entirely, but they reduce how easily another researcher could:

- independently interpret the dataset,
- reproduce portions of the analytical workflow,
- or fully evaluate the published findings using only the repository materials.

The assessment therefore concluded that the dataset supports partial reproducibility but would benefit from stronger standalone documentation.

Recommendations for Improvement

The following improvements would strengthen reproducibility support:

- provide a detailed data dictionary,
- document preprocessing and transformation steps,
- include clearer scoring explanations,
- expand README documentation,
- and add workflow transparency notes connecting the dataset with the analytical methods described in the article.